

Requirement Engineering and CMM Levels



Requirement Engineering at Level 1 - Initial

Characteristics

- Processes are chaotic, reactive and unstructured.
- Success depends on individual efforts, not processes.

Practices

- Informal or absent.
- Requirements are often misunderstood or incomplete.
- Changes lead to confusion and rework.

RC → Rework

CMM
↓
RE
↓
Import

Confidence
↓
specific REA
Maturity
Model

Requirement Engineering at Level 2-Repeatabile

Characteristics

- Basic Project management practices are established.
- Focus on managing requirements and schedules

Practices

- Requirements are formally documented.
- Baselines are established to control scope.
- Change control is introduced to manage otifications
- Stakeholder involvement become a priority.

✓ DMADIV
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→ 35 Step Req

CurA - Req 3
↓

Ni -
limited

→ Scope
→ Sch 1 h

Custom
= 8y R

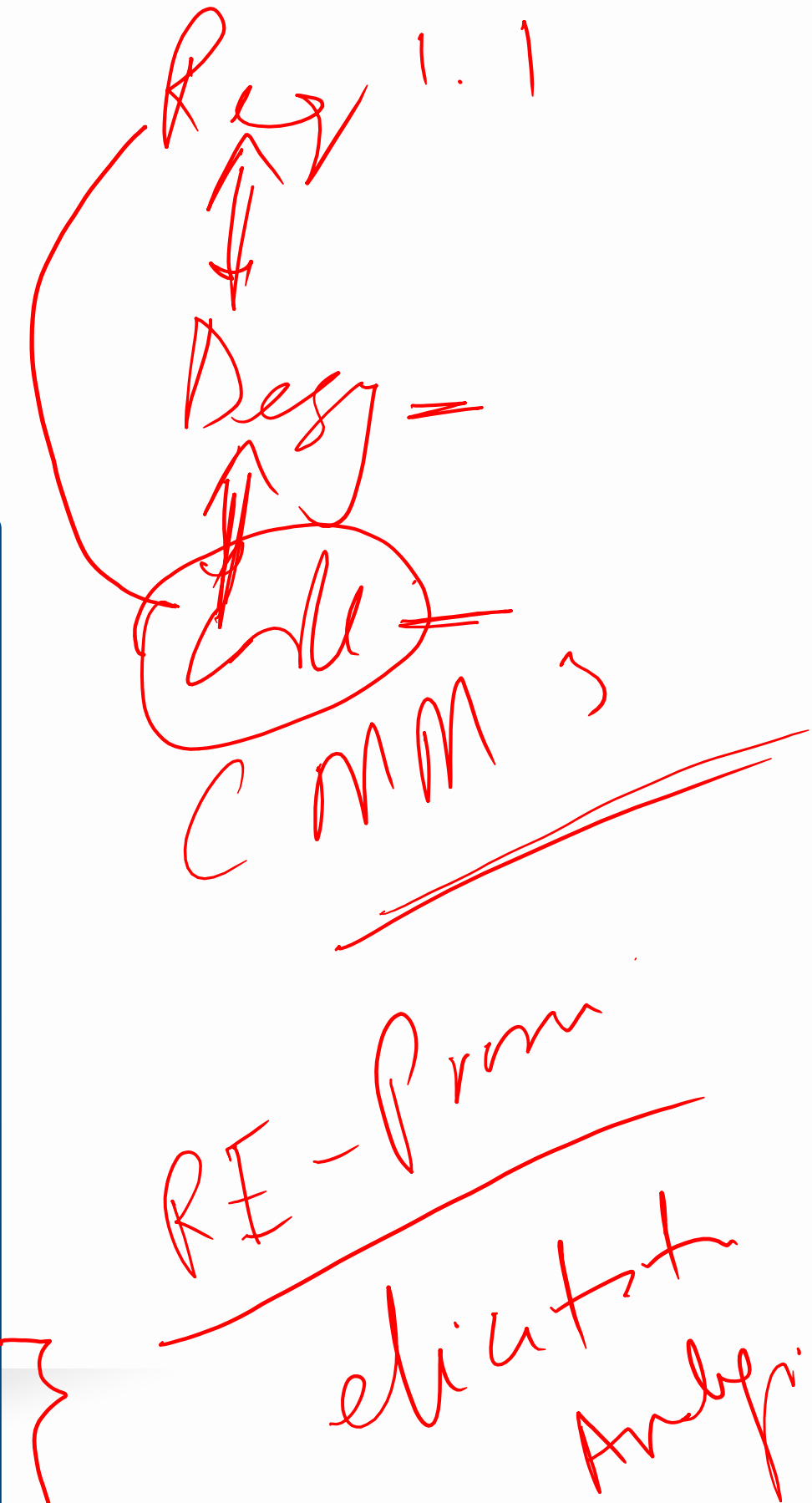
Requirement Engineering at Level 3 - Defined

Characteristics

- Processes are standardized and documented across the organization.
- A well-defined Requirements Development Process is implemented.

Practices

- Structured techniques like stakeholder interviews and surveys.
- Use cases, prototyping, and mockups clarify requirements.
- Identification of both functional and non-functional requirements.
- Traceability is maintained across requirements, design, and testing phases.



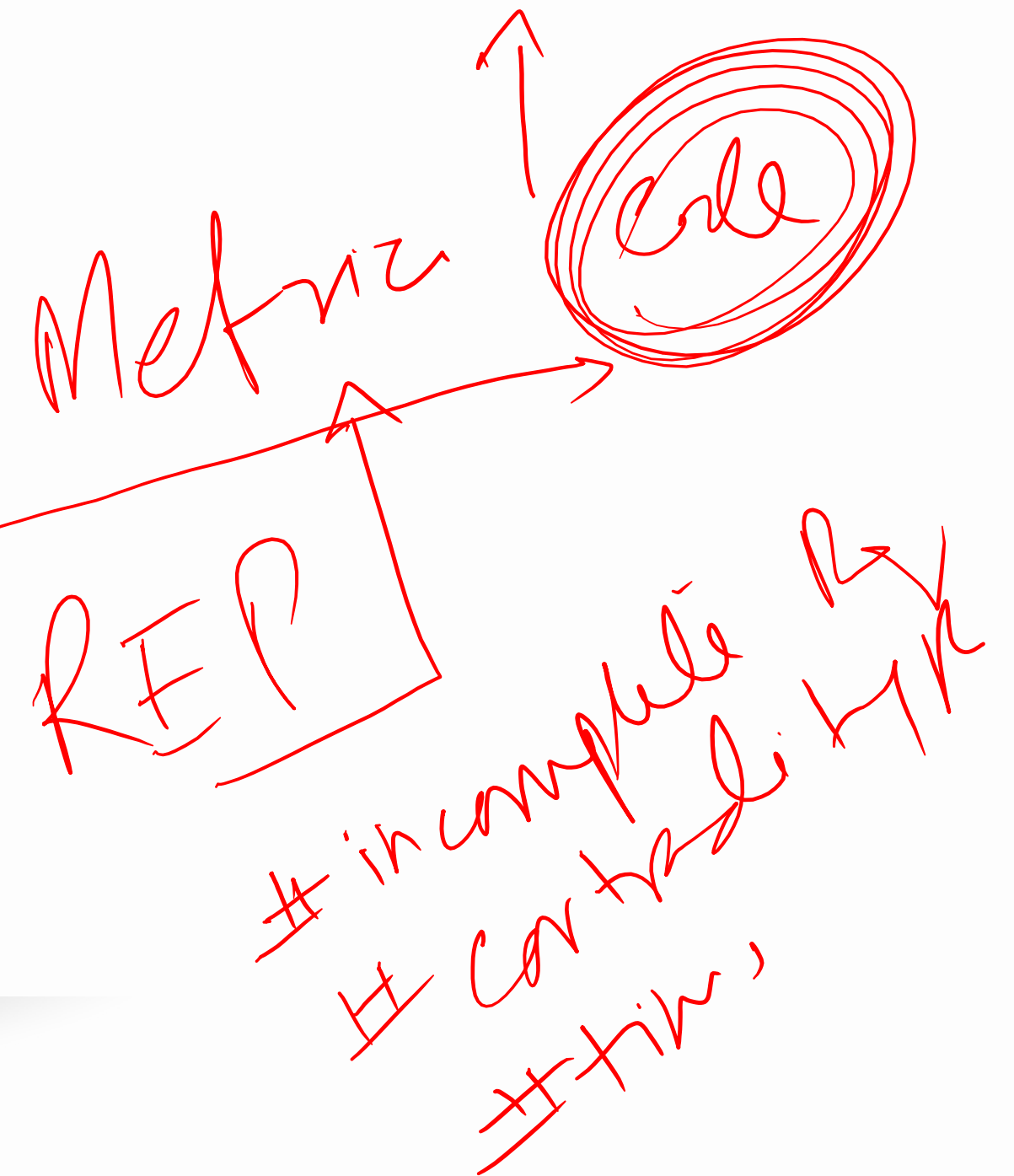
Requirement Engineering at Level 4 - Managed

Characteristics

- Quantitative metrics are used to monitor and control processes.
- Requirements processes are predictable and stable.

Practices

- Metrics to monitor requirements quality and stability, such as:
- Requirements Stability Index.
- Tracking requirement-related defects.
- Advanced tools for traceability and impact analysis of changes.
- Predictive analysis for assessing potential risks in requirements.



Requirement Engineering at Level 5 - Optimizing

Characteristics

- Processes are continuously improved using feedback and innovation.
- Focus on defect prevention and process optimization.

Practices

- Lessons learned are incorporated into refining processes.
- Implementation of advanced techniques, including:
- AI-powered tools for gathering and analyzing requirements.
- Simulations and predictive modeling for validation.
- Regular reviews of metrics and processes to identify

